Sina Malakouti

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Legal status in US: Permanent Resident

EDUCATION

University of Pittsburgh, Pittsburgh, PA **2020 - Present**
PhD in Computer Science
- Advisor: Seong Jae Hwang

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran **2015 - 2020**
B.Sc. in Software Engineering
- GPA: 17.47/20 (3.72/4)

FIELDS OF INTEREST

Machine Learning, Deep Learning, Computer Vision

TECHNICAL SKILLS

PROGRAMMING LANGUAGE
Java, Python, C/C++

WEB PROGRAMMING
Node.js, Flask, JETTY, JavaScript, jQuery, Vue.js, HTML, CSS, PHP

SMARTPHONE APP DEVELOPMENT
ReactNative, Objective C

DATABASE
MySQL, MongoDB, SQLite

MACHINE LEARNING & DATA SCIENCE
MATLAB, R, TensorFlow, Deeplearning4j, WEKA, scikit-learn, Pandas, NumPy

OPERATING SYSTEM
Linux, Mac OS X, Windows

RESEARCH EXPERIENCE

Research Assistant Intern, Mainz, Germany (July - Oct 2018)
- Data Mining Lab, Johannes Gutenberg University

The aim of this project was designing a new symbolic deep architecture based on Hoeffding trees which we **published in ECML PKDD 2019**.

Research Assistant Intern, Tehran, Iran (June – Sep 2019)
- Center for Biomedical and Health Informatics, Shahid Rajaei Cardiovascular Hospital, Medical Research Center

Applying Machine Learning to solve biomedicals problems, in particular, cardiovascular diseases.

Undergraduate Research Assistant, Tehran, Iran (June 2017 – May 2020)
- Big Data Lab, Amirkabir University of Technology

Working on Ensemble and merging of Decision Trees based on data distribution and Density Estimation in order to achieve a single model.

WORK EXPERIENCE

Software Engineer, Papillon (June – Dec 2016)
Papillon is a recipe social media for Iranian users with more than 1,000,000 users. My team and I worked on developing this application for Apple product users.

PUBLICATIONS

Ahamdi, Sina Malakouti, Kramer. "DeepTreeNetworks: A New Symbolic Deep Architecture." *Deep Continuous-Discrete Machine Learning (DeCoDeML)* ***ECML/PKDD 2019*** *workshop* (paper)

SELECTED OPEN SOURCE AND COURSE PROJECTS

Human face recognition using Eigen-face method

Classica; & Modern, Local Search Package - GITHUB

Inverted Index Search, tree structures, and hash table data structure library - GITHUB

Full stack online food ordering website - GITHUB

Multi-threading with AI 2D graphical strategic game - GITHUB

PRESENTATIONS

- A New Symbolic Deep Architecture." *Deep Continuous-Discrete Machine Learning (DeCoDeML)* ***ECML/PKDD 2019*** *workshop*, Würzburg, Germany

TEACHING EXPERIENCE

Teaching Assistant

Data Mining, Linear Algebra, Probabilities & Statistics, Design of Algorithms, Data Structures, Database Design, Web Programming, Java

RELATED COURSES

University Courses

Data Mining, Probabilities & Statistics, Linear Algebra, Signals & Systems, Stochastic Processes, Artificial Intelligence, Algorithms, Data Structures, Database Design, and Discrete Mathematics.

Online Courses

Deep Learning, Machine Learning, Algorithm Specialization, Linear Algebra, Basic Statistic

EXTRACURRICULAR & LEADERSHIP ACTIVITIES

Elected as the President of Student Scientific Chapter of Computer Engineering Department, Amirkabir University of Technology – (Jan 2017- March 2018)

My team and I held more than 70 national and international events, contests, seminars, and symposiums.

- Executive Director of the 16th Amirkabir International Collegiate Programming Contest (ACM ICPC) jointly with KTH – Royal Institute of Technology & TUM- Technical University of Munich.

HONORS & AWARDS

- Selected as an Outstanding Student (GPA based) and eligible to choose Second Major, Amirkabir University of Technology

LANGUAGES

English (TOEFL iBT Score: 106/ 120), **Farsi** (Native)

REFERENCES

- **Dr. Seong Jae Hwang**, University of Pittsburgh: sjh94@pitt.edu
- **Dr. Maryam Amir Haeri**, Amirkabir University of Technology: haeri@aut.ac.ir
- **Prof. Stefan Kramer**, Johannes Gutenberg University: kramer@informatik.uni-mainz.de
- **Zahra Ahmadi**, Johannes Gutenberg University: zaahmadi@informatik.uni-mainz.de